

Homework 1

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Claim 3a. *Show that the composition of homotopy equivalences $X \rightarrow Y$ and $Y \rightarrow Z$ is a homotopy equivalence $X \rightarrow Z$. Deduce that homotopy equivalence is an equivalence relation.*

Proof. Transitive: Suppose $X \simeq Y$ and $Y \simeq Z$. Thus there are functions f_1, f_2, g_1, g_2 where

$$X \xrightarrow{f_1} Y \xrightarrow{f_2} Z$$

$$X \xleftarrow{g_1} Y \xleftarrow{g_2} Z$$

where $f_1 g_1, g_1 f_1, f_2 g_2$, and $g_2 f_2$ are homotopic to the identity maps in their relevant spaces.

Let $f := f_2 f_1$ and let $g = g_1 g_2$. Thus let F_1 and G_1 be the homotopies that give $g_2 f_2 \simeq 1_Y$ and $g_1 f_1 \simeq 1_X$. Let $H : X \times [0, 1] \rightarrow X$ be the following:

$$H(x, t) := \begin{cases} g_1(F_1(f_1(x), 2t)) & \text{if } t \in [0, \frac{1}{2}] \\ G_1(x, 2t - 1) & \text{if } t \in [\frac{1}{2}, 1] \end{cases}$$

$g_1(F_1(f_1(x), 2t))$ is continuous on $[0, \frac{1}{2}]$ and $G_1(x, 2t - 1)$ is continuous on $[\frac{1}{2}, 1]$. Furthermore at $t = \frac{1}{2}$, $F_1(f_1(x), 2t) = F_1(f_1(x), 1) = (1_Y \circ f_1)(x) = f_1(x) \implies g_1(F_1(f_1(x), 2t)) = g_1(f_1(x))$ at $t = \frac{1}{2}$. Similarly at $t = \frac{1}{2}$, $G_1(x, 2t - 1) = (g_1 \circ f_1)(x)$. Thus H is well defined and is continuous by the Pasting Lemma. Furthermore at $H(x, 0) = g_1(F_1(f_1(x), 0)) = g_1(g_2(f_2(f_1(x)))) = (g \circ f)(x)$ and $H(x, 1) = G_1(x, 1) = 1_X$. Thus $g \circ f \simeq 1_X$.

Let F_2 and G_2 be the homotopies that give you $f_1 g_1 \simeq 1_Y$ and $f_2 g_2 \simeq 1_Z$. Let $K : Z \times [0, 1] \rightarrow Z$ be the following:

$$K(z, t) := \begin{cases} f_2(F_2(g_2(z), 2t)) & \text{if } t \in [0, \frac{1}{2}] \\ G_2(z, 2t - 1) & \text{if } t \in [\frac{1}{2}, 1] \end{cases}$$

$f_2(F_2(g_2(z), 2t))$ is continuous on $[0, \frac{1}{2}]$ and $G_2(z, 2t - 1)$ is continuous on $[\frac{1}{2}, 1]$. Furthermore at $t = \frac{1}{2}$, $F_2(g_2(z), 2t) = F_2(g_2(z), 1) = (1_Y \circ g_2)(z) = g_2(z) \implies f_2(F_2(g_2(z), 2t)) = f_2(g_2(z))$ at $t = \frac{1}{2}$. Similarly at $t = \frac{1}{2}$, $G_2(z, 2t - 1) = (g_2 \circ f_2)(z)$. Thus K is well defined and is continuous by the Pasting Lemma. Furthermore at $K(z, 0) = f_2(F_2(g_2(z), 2t)) = (f_2 \circ f_1 \circ g_1 \circ g_2)(z) = (f \circ g)(z)$ and $K(z, 1) = G_2(z, 1) = 1_Z$. Thus $f \circ g \simeq 1_Z$.

Thus $X \simeq Z$.

Symmetric: Suppose $f : X \rightarrow Y$ is a homotopy equivalence. Thus there is a function $g : Y \rightarrow X$ such that $f g \simeq 1_Y$ and $g f \simeq 1_X$. Thus $g : Y \rightarrow X$ is a homotopy equivalence. Thus $Y \simeq X$.

Reflexive: Let $F(x, t) = x = (1_X \circ 1_X)(x) = 1_X(x)$, F is continuous. Thus $1_X \circ 1_X \simeq 1_X$. Thus $X \simeq X$.

Thus homotopy equivalences are equivalence relations. $\square$

Claim 3b. *Show the relation of homotopy is an equivalence relation*

Proof. 1. **Reflexive:** Let $f : X \rightarrow Y$. Then let $F : X \times [0, 1] \rightarrow Y$ where $(x, t) \rightarrow f(x)$. For all open sets $U \subseteq Y$, $F^{-1}(U) = \{(x, t) \in X \times I \mid f(x) \in U\} = f^{-1}(U) \times I$ which is open. Thus F is continuous. $F(x, 0) = f(x)$ and $F(x, 1) = f(x)$. Thus $f \simeq f$.

2. **Symmetric:** Let $f : X \rightarrow Y$ and $g : X \rightarrow Y$ where $f \simeq g$. Thus $\exists F : X \times [0, 1] \rightarrow Y \ni F(x, 0) = f(x), F(x, 1) = g(x)$. Let $G : X \times I \rightarrow Y \ni G(x, t) = F(x, 1 - t)$ now $G(x, 0) = g(x)$ and $G(x, 1) = f(x)$. Note $(x, t) \rightarrow (x, 1 - t)$ is continuous because its identity map (which is continuous) on X and a linear map (which is continuous) on I . Thus since G is a composition of F with $(x, t) \rightarrow (x, 1 - t)$, it is continuous. Thus $g \simeq f$.

3. **Transitive:** Suppose $f \simeq g$ and $g \simeq h$. Let F and G be the homotopies that give you $f \simeq g$ and $g \simeq h$. $H : X \times I \rightarrow Y$ where

$$H(x, t) = \begin{cases} F(x, 2t) & \text{if } x \in [0, \frac{1}{2}] \\ G(x, 2t - 1) & \text{if } x \in [\frac{1}{2}, 1] \end{cases}$$

$$g(x) = G(x, 0) = H\left(x, \frac{1}{2}\right) = F(x, 1) = f(x)$$

Thus H is well defined and by the pasting lemma it is also continuous. $H(x, 0) = F(x, 0) = f(x)$ and $H(x, 1) = G(x, 1) = h(x)$. Thus $f \simeq h$. $\square$

Claim 3c. *Show that a map homotopic to a homotopy equivalence is a homotopy equivalence*

Proof. Suppose $f : X \rightarrow Y$ is a homotopy equivalence with $g : Y \rightarrow X$ be its corresponding homotopy equivalence from Y to X . Suppose $\alpha \simeq f$. Let F_1, F_2 be the homotopies which give $g \circ f \simeq 1_X$ and $f \circ g \simeq 1_Y$ respectively. Let G be the homotopy which gives $\alpha \simeq f$. Let $H : X \times [0, 1] \rightarrow X$ where

$$H(x, t) = \begin{cases} g(G(x, 2t)) & \text{if } x \in [0, \frac{1}{2}] \\ F_1(x, 2t - 1) & \text{if } x \in [\frac{1}{2}, 1] \end{cases}$$

$g(G(x, 2t))$ is continuous because g , G , and $[(x, t) \rightarrow (x, 2t)]$ are continuous and $g(G(x, 2t))$ is a composition of the three. $[(x, t) \rightarrow (x, 2t - 1)]$ is continuous thus $F_1(x, 2t - 1)$ is continuous.

$$g(f(x)) = g(G(x, 1)) = H\left(x, \frac{1}{2}\right) = F_1(x, 0) = g(f(x))$$

Thus H is well defined and by the pasting lemma it is continuous. Furthermore, $G(x, 0) = \alpha(x) \implies H(x, 0) = g(\alpha(x))$ and $F_1(x, 1) = 1_X \implies H(x, 1) = 1_X$. Thus $g \circ \alpha \simeq 1_X$

Let $K : Y \times I \rightarrow Y$ where

$$K(x, t) = \begin{cases} G(g(y), 2t) & \text{if } x \in [0, \frac{1}{2}] \\ F_2(y, 2t - 1) & \text{if } x \in [\frac{1}{2}, 1] \end{cases}$$

Since g is continuous, $[(y, t) \rightarrow (g(y), 2t)]$ is continuous. Thus $G(g(y), 2t)$ is continuous because g , G , and $[(y, t) \rightarrow (g(y), 2t)]$ are continuous and $G(g(y), 2t)$ is a composition of the three. $[(y, t) \rightarrow (y, 2t-1)]$ is continuous thus $F_2(y, 2t-1)$ is continuous.

$$f(g(y)) = G(g(y), 1) = K\left(y, \frac{1}{2}\right) = F_2(y, 0) = f(g(y))$$

Thus Y is well defined and by the pasting lemma it is continuous. Furthermore, $G(x, 0) = \alpha(x) \implies K(y, 0) = \alpha(g(y))$ and $F_2(y, 1) = 1_Y \implies H(y, 1) = 1_Y$. Thus $\alpha g \simeq 1_Y$.

Thus α is a homotopy equivalence. □

Claim 5. *Show that if a space X is a deformation retracts to a point $x_0 \in X$, then for each neighborhood U of x_0 in X there exists a neighborhood $V \subseteq U$ of x_0 such that $V \hookrightarrow U$ is nullhomotopic.*

Proof. Fix a neighborhood U of x_0 . Since $1_X \simeq [x \rightarrow x_0] \implies \exists F : X \times I \rightarrow X \ni F(x, 0) = 1_X(x) = x, F(x, 1) = x_0, F(x, t) = x_0$ for all $t \in I$. Since F is a deformation retraction, $F(x_0 \times I) = \{x_0\} \subseteq U \implies x_0 \times I \subseteq F^{-1}(U)$. Since I is compact, by the tube lemma $\exists V \subseteq X$ a neighborhood of x_0 such that $x_0 \times I \subseteq V \times I \subseteq F^{-1}(U)$. Let $i : V \hookrightarrow U$ be the natural inclusion map. Let $G : V \times I \rightarrow U \ni (x, t) \rightarrow F(x, t)$. Since $\forall (v, t) \in V \times I, F(v, t) \in U$, G is a valid function. $G(x, 0) = F(x, 0) = 1_X(x) = i(x)$ and $G(x, 1) = x_0$ and since $G = F|_{V \times I}$ G is continuous. Thus $i \simeq [v \rightarrow x_0]$ and thus is nullhomotopic. □

Problem 6a. *Let X be the subspace of $\mathbb{R}^2$ consisting of the horizontal segment $[0, 1] \times \{0\}$ together with the vertical segments $\{r\} \times [0, 1-t]$ for $r \in \mathbb{Q} \cap [0, 1]$. Show X deformation retracts to any point in the segment $[0, 1] \times \{0\}$ but not any other point.*

Solution. For all $x_0 \in [0, 1]$. Let $G : X \times I \rightarrow X$ where $((x, y), t) \rightarrow (x, (1-t)y)$ and is continuous. Thus $G(\vec{x}, 0) = \vec{x}$ and $G((x, y), 1) = (x, 0)$. Let $F : X \times I \rightarrow X$ where

$$F((x, y), t) = \begin{cases} G((x, y), 2t) & \text{if } x \in [0, \frac{1}{2}] \\ (x + (x_0 - x)(2t - 1), 0) & \text{if } x \in [\frac{1}{2}, 1] \end{cases}$$

$G((x, y), 2t)$ is continuous and $[(x, y), t) \rightarrow (x + (x_0 - x)(2t - 1), y)]$ is continuous as well on $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ respectively.

$$(x, 0) = G((x, y), 0) = F\left(x, \frac{1}{2}\right) = (x + (x_0 - x)(0), y) = (x, 0)$$

Thus F is well defined and by the pasting lemma its continuous. $F((x, y), 0) = G((x, y), 0) = 1_X(x, y)$ and $F((x, y), 1) = (x + (x_0 - x), 0) = (x_0, 0)$. $F((x, y), 1) = (x_0, 0)$ and $\forall t \in [0, \frac{1}{2}] F((x_0, 0), t) = (x_0, t)$ and $\forall t \in [\frac{1}{2}, 1], F((x_0, 0), t) = (x_0 + 0, 0) = (x_0, 0)$. Thus F is a deformation retraction.

$\forall r \in \mathbb{Q} \cap [0, 1] \forall s \in [0, 1-r]$, let $x_0 = (r, s)$. Let $J = [0, 1] \times \{0\}$. Let $U = X \setminus J$, U is open and $x_0 \in U$. Note X is path connected, however U is not and the path components of U are $U_q := \{q\} \times (0, 1-q]$ for $q \in \mathbb{Q} \cap [0, 1]$. For any open set $V \subseteq U$, V can be thought of as the union of the intersection of a collection of open balls with U disjoint from J , and thus V intersects X at a countably infinite number of disjoint line segments and thus V is not path connected. Assume the contrary that X deformation retracts to x_0 . Thus by Claim 5, $\exists V \subseteq U$ a neighborhood of x_0 such that the inclusion $i : V \hookrightarrow U$

is nullhomotopic. Let $F : V \times I \rightarrow \{x_0\}$ be the corresponding homotopy where $F(x, 0) = 1_V(x)$ and $F(x, 1) = x_0$. Thus $\forall x \in V$ there is a path from $x \rightarrow x_0$ where $t \rightarrow f_t(x)$. Thus V is path connected which is a contradiction. Thus by contradiction, X cannot have a deformation retraction to a point in $X \setminus J$. ■

Claim 9. *Show that a retract of a contractible space is contractible.*

Proof. Suppose X is a contractible space and suppose $A \subseteq X$ is a retract of X , and let r be the corresponding retraction. Since X is contractible, $X \simeq \{x_0\}$ for some $x_0 \in X$. Thus $\exists f : X \times I \rightarrow \{x_0\}$ where $x \rightarrow x_0$ and $\exists g : \{x_0\} \times I \rightarrow X$ where $gf \simeq 1_X$ and $fg \simeq 1_{\{x_0\}}$. Let $i : A \hookrightarrow X$ be the natural inclusion.

$$A \xhookrightarrow{i} X \xrightarrow{f} \{x_0\}$$

$$A \xleftarrow{r} X \xleftarrow{g} \{x_0\}$$

Let $\alpha := f \circ i$ and $\beta := r \circ g$. Let F be the homotopy that gives you $gf \simeq 1_X$ and let G be the homotopy that gives you $fg \simeq 1_{\{x_0\}}$. I will omit i from now on and substitute $i(a)$ with a .

Let $H : A \times I \rightarrow A$ be $H(a, t) = r(F(a, t)) = (r \circ F)(a, t)$ and thus H is continuous. $H(a, 0) = r(F(a, 0)) = r(g(f(a))) = r(g(f(i(a)))) = (rgfi)(a) = (\beta \circ \alpha)(a)$. $H(a, 1) = r(1_X(a)) = r(a) = 1_A(a) = a$. Thus $\beta\alpha \simeq 1_X$.

Let $K : \{x_0\} \times I \rightarrow \{x_0\}$ where $K(x_0, t) = (\alpha\beta)(x_0) = f(i(r(g(x_0)))) = x_0 = 1_X$, thus $\alpha\beta = 1_X \implies \alpha\beta \simeq 1_X$.

Thus $A \simeq \{x_0\} \implies A$ is contractible. □

Claim 10a. *Show that a space X is contractible iff every map $f : X \rightarrow Y$, for arbitrary Y , is nullhomotopic.*

Proof. $\implies$: Suppose X is contractible. Fix an arbitrary topological space Y . Suppose $f : X \rightarrow Y$ is a continuous function. $\forall y_0 \in Y$, by definition of contractible $X \simeq \{y_0\}$. Thus $g : X \rightarrow \{y_0\}$ and $h : \{y_0\} \rightarrow X$ are functions such that $hg \simeq 1_X$ and $gh \simeq 1_{\{y_0\}}$. Note that $\forall x \in X, (h \circ g)(x) = h(y_0)$. Let $F : X \times [0, 1] \rightarrow X$ be the homotopy that gives you $hg \simeq 1_X$. Let $G : X \times [0, 1] \rightarrow Y$ where $x \rightarrow (f \circ F)(x, t)$ which is continuous. $\forall x \in X, G(x, 0) = f(F(x, 0)) = f(h(g(x))) = f(h(y_0)) \implies G(\cdot, 0) = [x \rightarrow f(h(y_0))]$ (ie a constant function). $\forall x \in X, G(x, 1) = f(F(x, 1)) = f(1_X(x)) = f(x) \implies G(\cdot, 1) = f$. Thus $f \simeq [x \rightarrow f(h(y_0))]$, thus f nullhomotopic.

$\impliedby$: Suppose every map $X \rightarrow Y$, for an arbitrary Y , is nullhomotopic. Then 1_X is nullhomotopic, thus $1_X \simeq [x \rightarrow x_0]$ for some $x_0 \in X$. Then let $g : \{x_0\} \hookrightarrow X$ be the natural inclusion. Since $x_0 \in X, 1_{\{x_0\}} : X \rightarrow X$ by factoring through the inclusion, thus $1_{\{x_0\}} \simeq [x \rightarrow x_0]$. $g1_X \simeq [x \rightarrow x_0] \simeq 1_X \implies g1_X \simeq 1_X$ and since $1_X(g(x_0)) = x_0 \in X \implies 1_X \circ g = 1_{\{x_0\}} \implies 1_X \circ g \simeq 1_{\{x_0\}}$. Thus $X \simeq \{x_0\} \implies X$ is contractible. □

Claim 10b. *Show X is contractible iff every map $f : Y \rightarrow X$ is nullhomotopic.*

Proof. $\implies$: Suppose X is contractible. Fix an arbitrary topological space Y . Suppose $f : Y \rightarrow X$ is a continuous function. $\forall x_0 \in X$, by definition of contractible $X \simeq \{x_0\}$. Thus $g : X \rightarrow \{x_0\}$ and $h : \{x_0\} \rightarrow X$ are functions such that $hg \simeq 1_X$ and $gh \simeq 1_{\{x_0\}}$. Note that $\forall x \in X, (h \circ g)(x) = h(x_0) = x_1$.

Let $F : Y \times [0, 1] \rightarrow X$ be the homotopy that gives you $hg \simeq 1_X$. Let $G : X \times [0, 1] \rightarrow Y$ where $y \rightarrow F(f(y), t)$ which is continuous. $\forall y \in Y, G(y, 0) = F(f(y), 0) = h(g(f(y))) = x_1 \implies G(\cdot, 0) = [x \rightarrow x_1]$ (ie a constant function). $\forall y \in Y, G(y, 1) = F(f(y), 1) = 1_X(f(y)) = f(y) \implies G(\cdot, 1) = f$. Thus $f \simeq [y \rightarrow x_1]$, thus f nullhomotopic

$\Leftarrow$: Suppose every map $Y \rightarrow X$, for an arbitrary Y , is nullhomotopic. Then 1_X is nullhomotopic, thus $1_X \simeq [x \rightarrow x_0]$ for some $x_0 \in X$. Then let $g : \{x_0\} \hookrightarrow X$ be the natural inclusion. Since $x_0 \in X$, $1_{\{x_0\}} : \{x_0\} \rightarrow X$ by factoring through the inclusion, thus $1_{\{x_0\}} \simeq [x \rightarrow x_0]$. Since $g : \{x_0\} \rightarrow X \implies g \simeq [x \rightarrow x_0]$, $g1_X = g \simeq [x \rightarrow x_0] \simeq 1_X \implies g1_X \simeq 1_X$ and since $1_X(g(x_0)) = x_0 \in X \implies 1_X \circ g = 1_{\{x_0\}} \implies 1_X \circ g \simeq 1_{\{x_0\}}$. Thus $X \simeq \{x_0\} \implies X$ is contractible. $\square$

Claim 14. *Given positive integers v, e, f satisfying $v - e + f = 2$, construct a cell structure on S^2 having v 0-cells, e 1 cells, and f 2 cells*

Solution. My examples are the normal CW construction for S^2 with 2 0 cells, 2 1 cells, and 2 2 cells, a tetrahedron, a cube, and an octahedron as all CW constructions will create a CW Complex homeomorphic to S^2 . Other valid examples would be the other platonic solids as they are homeomorphic to S^2 and since they can be created into planar graphs they satisfy $v - e + f = 2$. $\blacksquare$

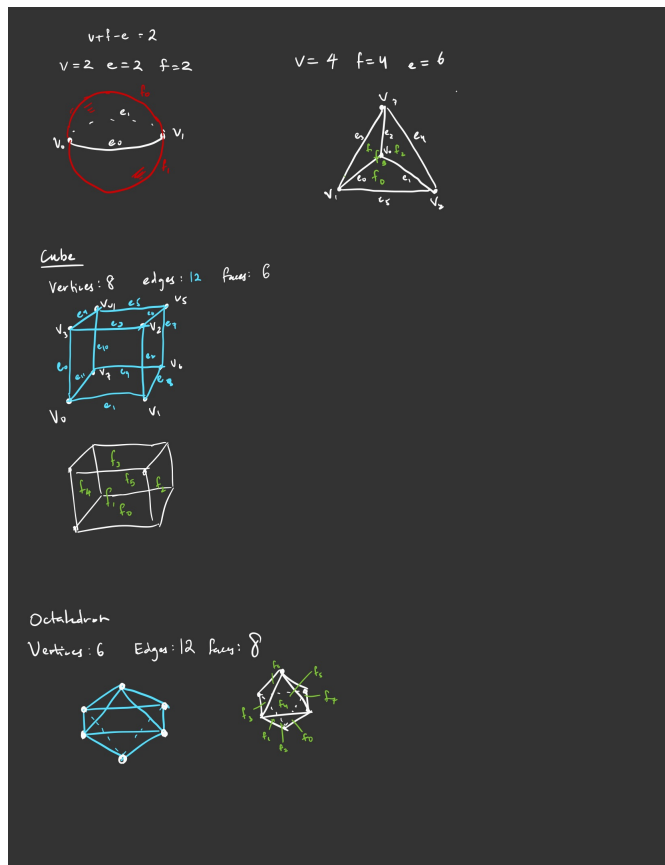


Figure 1: Your image caption

Claim 16. *Show that S^∞ is contractible*

Proof. Hatcher defines $S^\infty = \bigcup_{n \in \mathbb{N}}$ as the CW complex with two n -cells (corresponding to the northern / southern hemispheres of S^n) for each $n \geq 0$. Note for all $n \geq 1$, you can contract the n -cell corresponding to the northern hemisphere of S^n to x_0 , one of the base points of S^∞ , transforming S^n to S^n where S^n is created by a 0-cell and 1 n -cell.

Let $F_n : S^n \times I \rightarrow S^\infty$ be this homotopy. Since (S^∞, S^n) is a CW pair, homotopy extension property applies by Proposition 0.16. Thus there exists a homotopy $G_n : S^\infty \times I \rightarrow S^\infty$ which contracts S^n to x_0 . Note $G_n(\cdot, 0) = 1_{S^\infty}$. Also note $G_{n,t} := G_n(\cdot, t)$. Let $H : S^\infty \times I \rightarrow S^\infty$ where

$$H(x, t) := \begin{cases} G_1(x, 2t) & \text{if } t \in [0, \frac{1}{2}] \\ G_2(G_1(x, 1), 4t - 2) & \text{if } t \in [\frac{1}{2}, \frac{3}{4}] \\ G_3(G_2(G_1(x, 1), 1), 8t - 6) & \text{if } t \in [\frac{3}{4}, \frac{7}{8}] \\ \vdots & \\ G_n((G_{n-1,1} \circ G_{n-2,1} \circ \dots \circ G_{1,1})(x), 2^n(t - 1) + 2) & \text{if } t \in [1 - \frac{1}{2^{n-1}}, 1 - \frac{1}{2^n}] \\ \vdots & \\ x_0 & \text{if } t = 1 \end{cases}$$

Thus $H(\cdot, 0) = G_1(\cdot, 0) = 1_{S^\infty}$ and $H(\cdot, 1) = x_0$. Since S^∞ is given the weak topology, H is continuous iff $H|_{S^n \times I}$ is continuous for all n . This restriction is continuous because $H|_{S^n \times I}$ does the work of $G_1, \dots, G_n$, by composition, where for each $[1 - \frac{1}{2^{n-1}}, 1 - \frac{1}{2^n}]$ interval the starting position at that interval is the last position of the preceding interval. Then it stays stationary from $[1 - \frac{1}{2^n}, 1]$. By Pasting Lemma, $H|_{S^n \times I}$ is continuous. Thus H is continuous. Thus H is the homotopy which shows identity map is homotopic to constant map. Thus S^∞ is contractible. $\square$

Claim 17. *Show that the mapping cylinder of every map $f : S^1 \rightarrow S^1$ is a CW complex.*

Proof. For all continuous maps, $f : S^1 \rightarrow S^1$ let $M_f := (S^1 \times I) \sqcup_f S^1$. Here is the construction to create M_f as a CW-complex:

1. Fix some $x_0 \in S^1$, start with two points $X^0 := \{x_0, f(x_0)\}$.
2. Add three 1 cells as follows:
 - (a) Add a one cell with both endpoints on x_0 : e_1
 - (b) Add a one cell with both endpoints on $f(x_0)$: e_2
 - (c) Add a one cell with a endpoint at x_0 and a end point at $f(x_0)$: e_3
3. Add a two cell with top attached to e_1 , two sides (left and right) attached to e_3 , and the bottom side attached to e_2 based on the map $f : S^1 \rightarrow S^1$.

$\square$

Claim 17b.

Proof. There is a retraction from the annulus and the mobius strip onto their center circles S_A^1 and S_M^1 . Let r, s be the respective retractions. Let $A' := S_A^1 \sqcup_r A$ and $M' := S_M^1 \sqcup_r M$. Similar to a mapping cylinder, A' deformation retracts to S_A^1 and M' deformation retracts to S_M^1 . Now connect A' and M' with a cylinder to create the space X . So now A' is on one end of the cylinder and M' is on the other. To get the deformation retraction onto A , retract M' onto S_M^1 , retract the cylinder onto A' , then retract A' onto A . The process is identical for M , but switch A with M and A' with M' . $\square$

Claim 18a. *Show that $S^1 * S^1 = S^3$.*

Proof. On page 9 of Hatcher, $S^{n-1} \cong X_1 * \dots * X_n$ where $X_i \cong S^0$. Thus $S^1 \cong X_1 * X_2$. Thus $S^1 * S^1 = X_1 * X_2 * X_1 * X_2 = X_1 * X_2 * X_3 * X_4$ where $X_3 \cong X_1 \cong S^0$ and $X_4 \cong X_2 \cong S^0$. Thus $S^1 * S^1 = X_1 * X_2 * X_3 * X_4 = S^3$ $\square$

Claim 18b. *Show that $S^m * S^n = S^{m+n+1}$*

Proof. $S^{m+n+1} \cong X_1 * \dots * X_{m+n+2}$ where $X_i \cong S_0$.

Thus

$$\begin{aligned} S^{m+n+1} &\cong (X_1 * \dots * X_{m+1}) * (X_{m+2} * \dots * X_{m+n+2}) \\ &\cong S^m * S^n \end{aligned}$$

$\square$

Claim 21. *If X is a connected Hausdorff space that is a union of a finite number of 2 spheres, any two of which intersect in at most one point, show that X is homotopy equivalent to a wedge sum of S^1 's and S^2 's.*

Proof. This question is very similar to example 0.9 in Hatcher using a very similar algorithm. Base Case: Suppose X , connected Hausdorff space, is composed of two 2-spheres which intersect at at most 1 point. Since X is connected, the two 2-spheres must intersect at at least 1 point. Thus the two spheres intersect at 1 point. Thus by definition of wedge product. X is the wedge sum of S^2 's and thus is a wedge sum of S^1 's and S^2 's. Thus X is homotopy equivalent to a wedge sum of S^1 's and S^2 's.

Inductive Hypothesis: Assume that if X , is a connected Hausdorff space, and is a union of n 2 spheres, any two of which intersect in at most one point, then X is homotopy equivalent to a wedge sum of S^1 's and S^2 's.

Inductive Step: Let X , be a connected Hausdorff space, where X is a union of $n+1$ 2 spheres, any two of which intersect in at most one point. Take a arbitrary 2-sphere label it A . We want to show that $X \simeq A \vee Y$ where Y is a connected Hausdorff space and is a union of n 2 spheres, any two of which intersect in at most one point. Let $\{B_1, \dots, B_k\}$ be the spheres which intersect with A . Note there must be at least 1 since X is connected. Let $\{x_1, \dots, x_n\}$ be the points at which A and B_i intersect. We want to build intermediate spaces Z_i which are homotopy equivalent to X .

1. Let Z_1 be X .
2. Let $i = 2$.
3. While $i < k$ do the following:

- (a) If B_{i-1} and B_i in Z_{i-1} have a intersection. Figure 2 has the entire details of this section (any intersection other than $A.B_{i-1}$, and B_i is omitted as they don't matter in this stage.
) Create a string, 1 cell, between B_{i-1} and B_i where the endpoints, b_{i-1} and b_i , are the intersection point. Separate the intersection points, so they are no longer identified with each other. This is space W_i . $X \simeq W_i$.
- i. Then collapse the arcs of b_{i-1}, x_{i-1} (take $b_{i-1} \rightarrow x_{i-1}$) on B_{i-1} , x_{i-1}, x_i (take $x_{i-1} \rightarrow x_i$) on A , x_i, b_i (take $b_i \rightarrow x_i$) on B_i to get Z_i . Now B_{i-1}, A, B_i and a copy of S^1 all intersect at 1 point. Finally $X \simeq Z_i$. Since we collapsed the arcs of b_{i-1}, x_{i-1} in B_{i-1} and x_i, b_i and B_{i-1} , there is still only one intersection between B_{i-1} and B_i .
- (b) Otherwise, collapse the arc x_{i-1}, x_i (take $x_{i-1} \rightarrow x_i$) on A . Now B_{i-1}, A, B_i all intersect at 1 point. Finally $X \simeq Z_i$.

After this process, in space Z_k , A should intersect the other spheres at X at one point and only one point. Furthermore A and any of the S^1 's created in the previous algorithm intersects the rest of Z_k at only 1 point, x_k . Thus Z_k is a wedge sum of A (a copy of S^2), all the S^1 's created in the algorithm, and a space Y , a union of n 2-spheres where any two of which only intersect at 1 point. Y is created by removing A , all the copies of S^1 created in the algorithm, and adding the point x_k to ensure intersections aren't broken. By the inductive hypothesis, Y is a wedge sum of S^2 s and S^2 s. Thus $X \simeq Z_k$ and Z_k is homotopy equivalent to a wedge sum of S^2 s and S^2 s. Thus X is homotopy equivalent to a wedge sum of S^2 s and S^2 s.

Thus by induction for any finite n , X would be the wedge sum of S^1 s and S^2 s. □

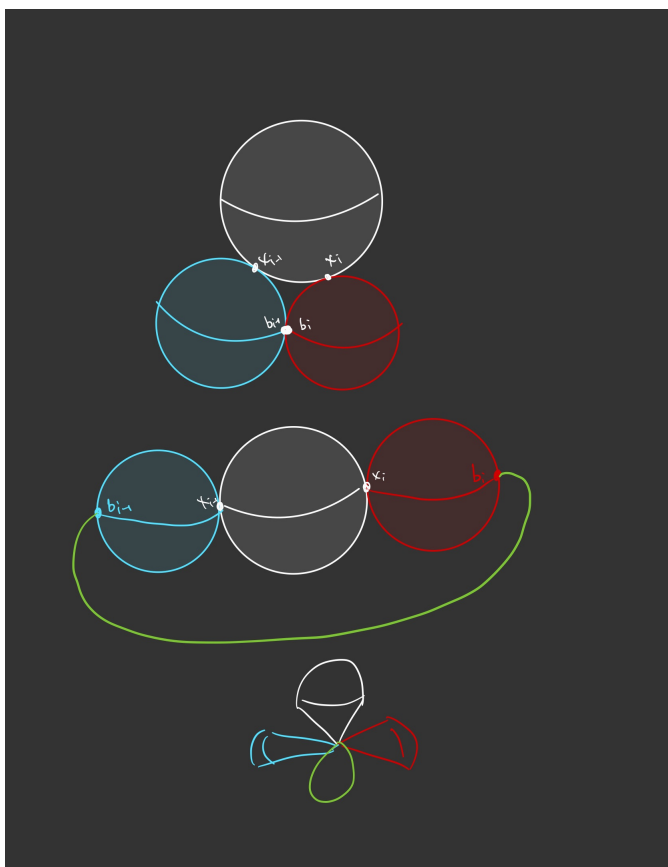


Figure 2: Section 3a

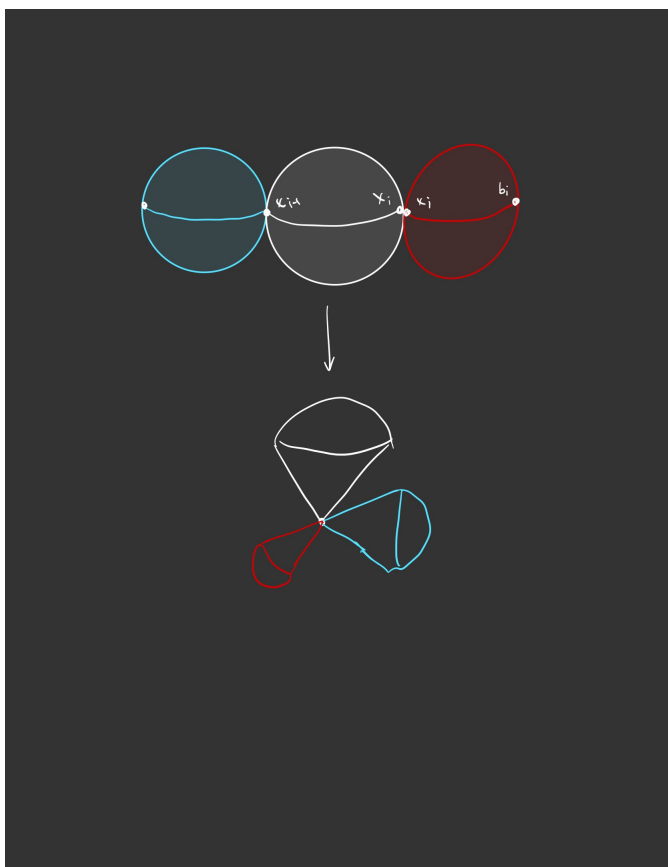


Figure 3: Your image caption